



EQUITY BANK PESALINK ISO20022 UPGRADE

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On-Premise

Pre-Deployment

Release Notes

Table Of Contents

Project Details	3
Release History	4
Confidentiality Disclaimer	5
Introduction	6
Production URLs And Endpoints	6
Mapping IPLS IP To Domain Name	6
Channel Certificates	7
Channel Certificates Renaming	7
Channel Certificates Deployment	7
Equity-IPSL CSR Certificate Generation	8
Definition Of Terms	8
CSR And Private Keys Generation Steps	8
Generating private key and CSR for Transport.....	8
IPSL Certificate Deployment	10

Project Details

Application Name	Equity Interbank
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Name of Re-leaser	Kennedy N. Gatimu
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Prepared By	Kennedy N. Gatimu

Release History

Date	Version	Author	Comments
12/09/2022	release-prod-1.0	Kennedy Gatimu	Initial Release

Confidentiality Disclaimer

DEVELOPER API GUIDE DOCUMENT

PROJECT: Equity Pesalink ISO20022 Upgrade

Date: 12.09.2022

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Introduction

This is the pre-deployment actions for Equity Pesalink ISO20022 upgrade on premise.
The actions include:

- I. Production urls and endpoints
- II. Mapping IPSL IP to domain name
- III. Channel Certificates
- IV. Equity - IPSL CSR generation and signing.

Production URLs And Endpoints

Below endpoints should be provided;

- I. AML Fraud Check Endpoint
- II. Endpoint To IPSL MGW Phone Lookup (ipsl.mgw.phone.lookup.url)
- III. IPSL Verification Request url (ips.verification.request.url)
- IV. IPSL Credit Transfer url (ips.credit.transfer.url)
- V. IPSL Payment Status Url (ips.payment.status.url)
- VI. Endpoint To IPSL MGW Phone Lookup (ipsl.mgw.phone.lookup.url)
- VII. Username To IPSL MGW Phone Lookup (kba.inquire.username)
- VIII. Password To IPSL MGW Phone Lookup (kba.inquire.password)
- IX. Production Microsoft SQL Database connection string and credentials

Mapping IPLS IP To Domain Name

From the production server;

1. Go to /etc/ and update the hosts file with the entry below

```
102.69.239.77 api.ips.pesalink.co.ke
```

Channel Certificates

Currently, we are only allowing below channel Ids with their respective source system Ids:

NO.	CHANNEL ID	SOURCE SYSTEM ID	SOURCE CHANNEL
1	APP	EQUITY_APP	OMNI
2	EVA	CHATBOT_EVA	
3	OMNI	OMNI	
4	USSD	EQUITY_USSD	
5	WEB	EQUITY_WEB	
6	STK	EQUITEL_STK	STK
7	REMITTANCE	REMITTANCE_INBOUND	GAG
8	OTC	BRANCH	FINACLE

Channel Certificates Renaming

- Omni Source Channel will issue one certificate.
 - Five (5) copies will be created from the original Omni certificate and named as;
 - EQUITY_APP.cer
 - CHATBOT_EVA.cer
 - OMNI.cer
 - EQUITY_USSD.cer
 - EQUITY_WEB.cer
- STK, GAG, and Finacle Source Channels will issue a certificate each.
 - For each, the certificate will be renamed, respectively, as;
 - EQUITEL_STK.cer
 - REMITTANCE.cer
 - BRANCH.cer

Channel Certificates Deployment

- Go to the base directory

```
cd /u01/gag/plv2/
```

- Create directory certs

```
mkdir certs
```

- From certs directory, create directory Equity-Channels

```
mkdir Equity-Channels
```

- Copy all the channel certificates to the Equity-Channels directory

```
cp EQUITY_APP.cer CHATBOT_EVA.cer OMNI.cer EQUITY_USSD.cer  
EQUITY_WEB.cer EQUITEL_STK.cer REMITTANCE.cer  
BRANCH.cer /u01/gag/plv2/certs/Equity-Channels
```

Equity-IPSL CSR Certificate Generation

Definition Of Terms

CSR - Certificate Signing Request

IPSL - Integrated Payment Systems Ltd

CSR And Private Keys Generation Steps

To connect to the new IPS, we need to generate two CSRs - Transport and Seal.
Below is the process for generating a private key and the CSR using openssl.

Generating private key and CSR for Transport

1. CSR Generation

Transport CSR Content

C=KE
ST=Nairobi
L=UPPERHILL
O=EQUITY BANK PLC
OU=PESALINK
emailAddress=info@equitybank.co.ke
CN=0068-transport.ips.pesalink.co.ke
subjectAltName = @alt_names
DNS.1=0068.ips.pesalink.co.ke

Transport CSR Generation

```
openssl req -newkey rsa:2048 -nodes  
-subj "/C=KE/ST=Nairobi/L=UPPERHILL/O=EQUITYBANK  
PLC/OU=PESALINK/emailAddress=info@equitybank.co.ke/CN=0068-transport.ips.  
pesal ink.co.ke/"  
-addext "subjectAltName=DNS.1:0068.ips.pesalink.co.ke"  
-keyout EQUITYBANK-transport.key  
-out EQUITYBANK-0068-transport.csr
```

Seal CSR Content

C=KE
ST=Nairobi
L=UPPERHILL
O=EQUITY BANK PLC
OU=PESALINK
emailAddress=info@equitybank.co.ke
CN=0068-seal.ips.pesalink.co.ke
subjectAltName = @alt_names
DNS.1=0068.ips.pesalink.co.ke

Transport CSR Generation

```
openssl req -newkey rsa:2048 -nodes  
-subj "/C=KE/ST=Nairobi/L=UPPERHILL/O=EQUITY BANK  
PLC/OU=PESALINK/emailAddress=info@equitybank.co.ke/CN=0068-seal.ips.pesalink.  
o.ke/"  
-addext "subjectAltName=DNS.1:0068.ips.pesalink.co.ke"  
-keyout EQUITYBANK-seal.key  
-out EQUITYBANK-0068-seal.csr
```

2. Send both transport and seal CSRs to IPLS for signing via email.

The email should have subject as;

[**ENVIRONMENT**]: [**PARTICIPANT CODE**] - [**CERT TYPE**] Certificate Signing Request:
DD MMM, YYYY

Example:

[**PRODUCTION**]: 0068-TRANSPORT Certificate Signing Request: 13th September, 2022
[**PRODUCTION**]: 0068-SEAL Certificate Signing Request: 13th September, 2022

3. Once the csrs are signed, the certificates will be shared on email as a securely zipped document containing .pem files. The password to open the document will be shared via the Whatsapp messaging app. Kindly share the contact (Mobile number) of the recipient from your side.

4. Using the .pem files, generate pkcs12 files for the respective pem files

```
I. openssl pkcs12 -export -in EQUITYBANK-0068-transport.pem -inkey  
EQUITYBANK-transport.key -name equitybanktransport -out  
EQUITYBANK_transport.p12
```

```
II. openssl pkcs12 -export -in EQUITYBANK-0068-seal.pem -inkey  
EQUITYBANK-seal.key -name equitybanktransport -out  
EQUITYBANK_seal.p12
```

Where **name** is the alias name.

3. Update cert password and share with dev team for code update.

This can be done using external tools like Portecle

7. Deploy the resulting .p12 file in the server.

IPSL Certificate Deployment

1. Go to the certificate base directory

```
cd /u01/gag/plv2/certs
```

2. From certs directory, create directory Equity-IPSL

```
mkdir Equity-IPSL
```

3. Copy both the transport and seal .p12 files to Equity-IPSL directory

```
cp EQUITYBANK_transport.p12 EQUITYBANK_seal.p12  
/u01/gag/plv2/certs/Equity-Channels
```